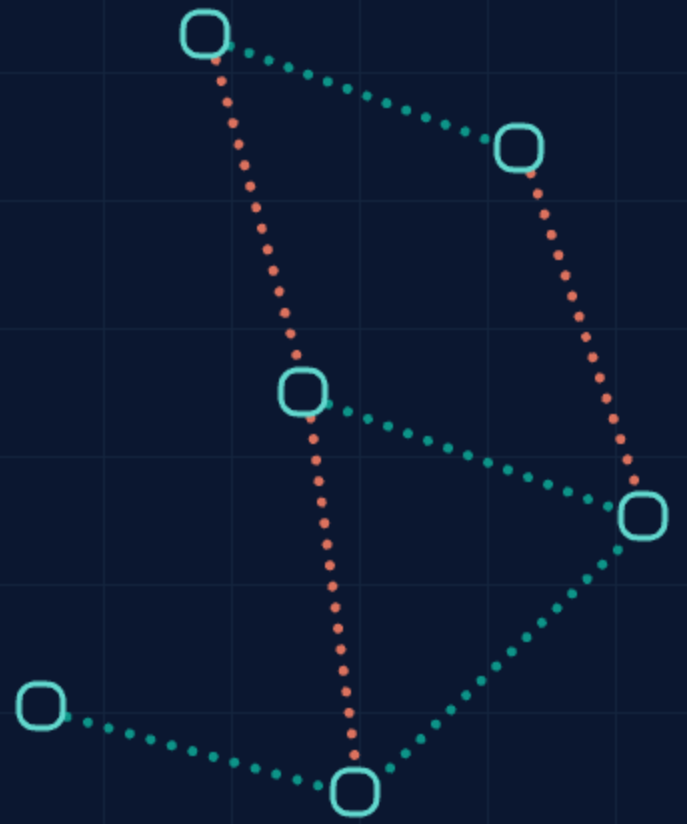
Q-FORCE / HACKATHON RESEARCH

QUANTUM RAILWAYS

Optimizing the Tracks of Tomorrow

Catalin · Artiom · Zus · Sabina · Livia



5 route decisions / 32 plans

Which connections should we build?

A fictional Romanian route network, five binary investment choices, one budget.

0	**Iași–Brașov**	cost 2 / score 4
1	**Iași–Cluj-Napoca**	cost 3 / score 6
2	**Cluj-Napoca–Brașov**	cost 2 / score 5
3	**Brașov–București**	cost 4 / score 9
4	**București–Constanța**	cost 4 / score 7

$x_i = 1$

choose route i

$\Sigma\ cost_i\ x_i \leq 7$

maximize $\Sigma\ score_i\ x_i$

Route costs and scores: railway_data.py; scenario route IDs 0, 1, 2, 6, 8.

32 plans. 16 valid. Two best.

Exhaustive enumeration is the ground truth for this five-route instance.

$2^5 = 32$

binary plans

16

budget-feasible

15 / 7

best score / cost

TIED OPTIMA

01010 routes 1 + 6

00111 routes 0 + 1 + 2

Exact enumeration on route IDs 0,1,2,6,8 · B=7 · score ties included.

A low-energy answer can break the budget

The squared penalty is centered on spending exactly B, while the rule allows spending less.

$$E(x) = -V(x) + \lambda\,[C(x) - B]^2$$

λ = 0.7

EXACT FEASIBLE OPTIMUM

score 15 · cost 7

Budget respected

EXACT QUBO MINIMUM

score 18 · cost 8

−17.3 energy · 1 over budget

Direct cost model, QUBO and Ising energies agree for all 32 states (max difference $<1.1\times10^{-14}$).

QAOA turns plan quality into interference

Five route decisions become five qubits; one layer has two trainable angles.

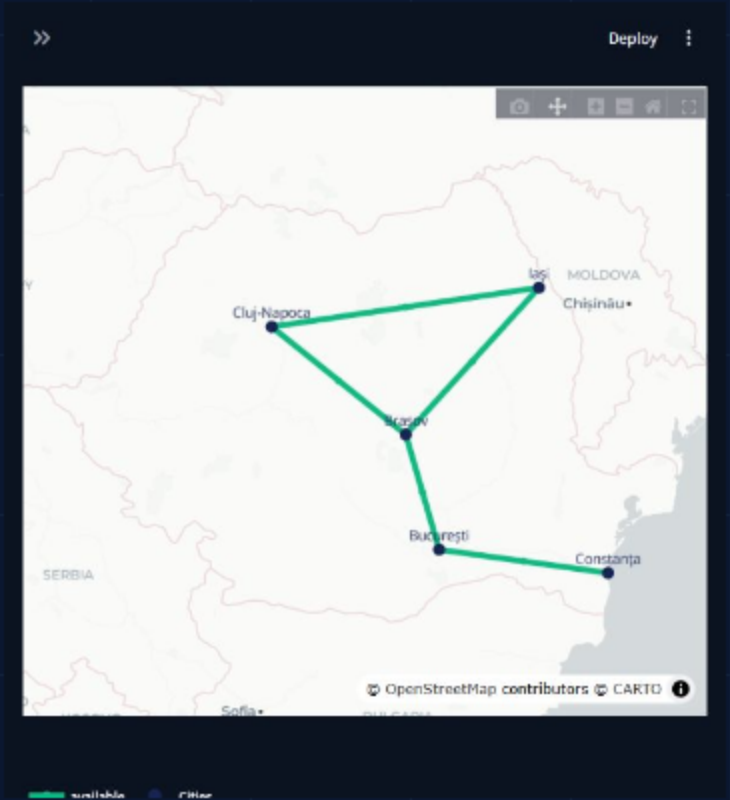


COBYLA adjusts γ and β to lower expected model energy; we then sample the trained circuit.

p=1 local training: ideal statevector expectation; ideal Aer and synthetic-noise Aer sampling.

An end-to-end planning laboratory

The app ties scenario design, exact reference, QAOA, maps and reproducible reports.



01 **Select routes & budget**

02 **Build QUBO + exact reference**

03 **Train and sample QAOA**

04 **Compare maps, statistics, reports**

Screenshot: running local Streamlit app, captured 8 Nov 2026; fictional schematic edges.

Change one axis; keep the comparison honest

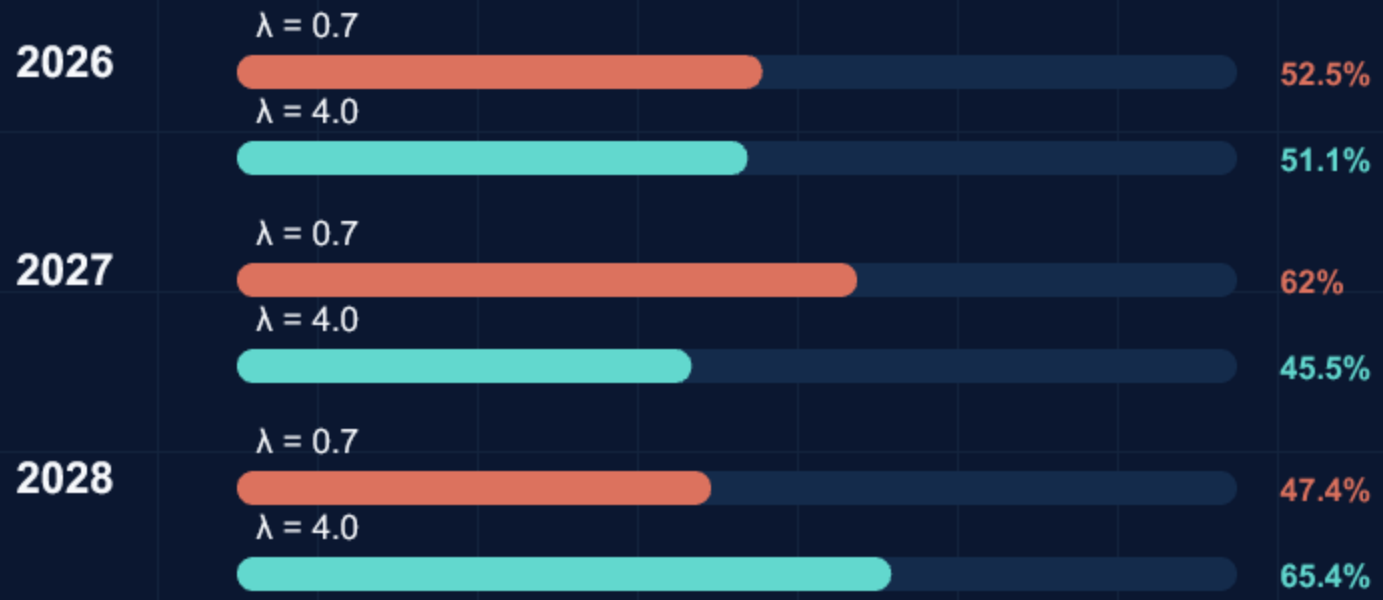
Run-level JSON preserves configurations, seeds, counts, resources and optimizer status.

PRIMARY	**Penalty λ**	0.7 vs 4.0	3 paired seeds
PRECISION	**Shots**	256–4096	fixed trained circuit
SCALE	**Route decisions**	5 → 7 → 8	$2^5 \rightarrow 2^8$ states
PLANNING	**Budget**	5 → 7 → 9	retrain per budget
EXECUTION	**Backend**	ideal vs local noise	same bound angles

Local replay: p=1, COBYLA, 30 evaluation limit, 2048 shots unless shot study; 8-route compare limit.

A stronger penalty fixes the model, not sampling

Exact validity flips at λ=4; measured feasible rates move both ways across seeds.



EXACT MODEL MINIMUM

λ .7 → invalid

λ 4 → valid

All six optimizers hit the evaluation limit.

Five routes, five cities, B=7, p=1, 2048 ideal shots, 30 evaluations, matched seeds.

More shots narrow uncertainty

We remeasured the same trained circuit; its underlying state did not change.

256	1024	2048	4096
shots	shots	shots	shots
~±6.1%	~±3.1%	~±2.2%	~±1.5%
95% sampling band	95% sampling band	95% sampling band	95% sampling band

Three seeds per shot count • fixed γ, β and scenario • finite-shot P(feasible) estimates

Wilson intervals from presentation/source_data/shot_sensitivity.json; band shown near p≈0.5.

Routes set qubits; cities do not

A route decision is one binary variable, so the state space grows as 2^n routes.

5 routes

5 cities

32 states

baseline

7 routes

7 cities

128 states

local compare

8 routes

7 cities

256 states

current compare limit

Budget 5 → 7 → 9

Exact best score 11 → 15 → 20

Size and budget replays: seed 2026, p=1, 2048 shots; each changed problem retrained.

The same angles, two simulators

One baseline run shows similar finite-shot feasibility; no noise benefit follows.

52.5%

ideal feasible

2048 shots

53.0%

synthetic noisy feasible

2048 shots

15

exact best feasible

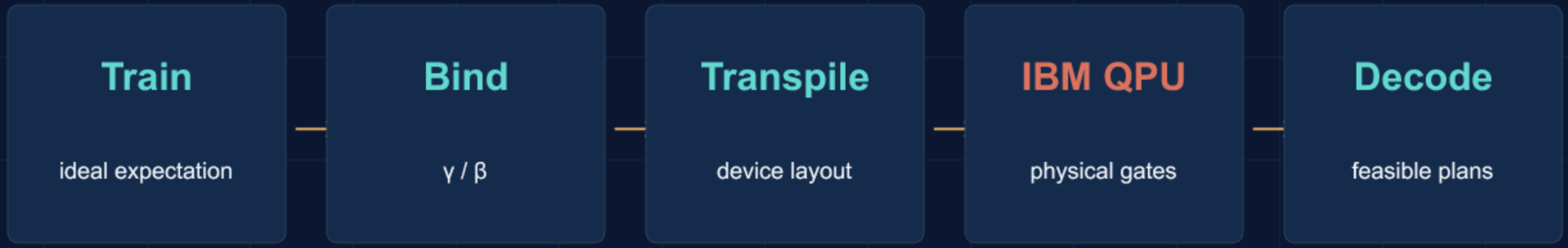
true score

Trained once → bound γ, β → sampled with Aer statevector and a local synthetic noise backend

Five routes, B=7, λ=.7, p=1, seed 2026; local GenericBackendV2 + Aer density matrix.

From trained angles to physical measurements

Hardware changes mapping, depth and error profile; decoding must follow the final layout.



Compare identical scenario, parameters, shots and decoded bit order wherever records permit.

Hardware workflow implemented separately in IBM runner; no verified counts available for this deck.

IBM QPU measurements remain unverified

The recorded job ID was not found in the configured IBM account at final check.

NO VERIFIED COUNTS

dbnpc00ou4as73b4u830

Recorded backend: ibm_marrakesh • 4096 requested shots • p = 1

Status query: RuntimeJobNotFound for the locally configured account

Recorded submission metadata only. No QPU success rate, score or distribution is claimed.

From a hackathon game to real Romania

A staged, source-grounded subnetwork study is the responsible next experiment.

01

CFR + ERA data

validate stations, tracks

02

Real network graph

connectivity & constraints

03

Public investment case

reported costs, transparent benefit

04

Classical benchmark

small regional subproblem

05

Quantum comparison

resources, validity, limits

CFR reports 10,615 km of lines and 896 operating points (2026 network statement annex).

CFR Network Statement 2026, Annex 5; ERA RINF Romania dataset. Real project costs not inferred from distance.

Meaningful solutions need more than low energy

Three lessons from one controlled railway optimization study.

01	**Exact classical reference**	reveals what is truly optimal
02	**Budget encoding**	can reward an invalid model minimum
03	**Training + sampling**	shape the plans we actually see

“Quantum optimization is about finding meaningful solutions.”

The implemented energy path

All 32 direct, QUBO and Ising energies matched numerically.

$$E(x) = -\Sigma\, v_i x_i + \lambda(\Sigma\, c_i x_i - B)^2$$

$$a_i = -v_i + \lambda(c_i^2 - 2Bc_i) \qquad b_{ij} = 2\lambda c_i c_j$$

$$x_i = (1 - Z_i)/2 \;\rightarrow\; H = \text{const} + \Sigma\, h_i Z_i + \Sigma\, J_{ij} Z_i Z_j$$

Direct/QUBO max $|\Delta E| = 1.07 \times 10^{-14}$; QUBO/Ising max $|\Delta E| = 7.11 \times 10^{-15}$.

Training stopped at its evaluation limit

A single archived 19-route depth test is illustrative, not a general depth result.

46

p = 1 observed best

19 routes, B=20

43

p = 2 observed best

same archived seed

171 → 342

two-qubit gates

p=1 → p=2

One archived seed (2026) • 30 evaluations each • neither optimizer converged

Tracked archive: validation/p_comparison_seed2026/comparison.json; configuration differs from five-route runs.

Every number has a source

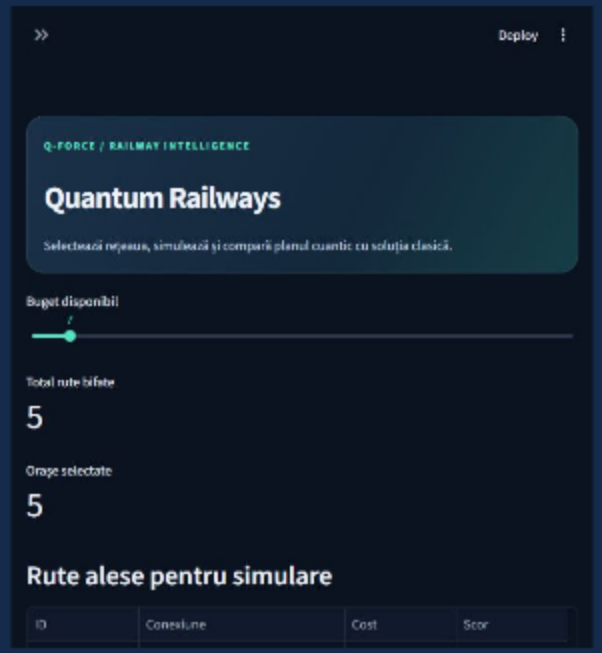
Rebuild from frozen run JSON; refresh hardware only after verified measurements.

LOCAL REPLAY	10 full run JSON files	presentation/experiments/
SHOT STUDY	12 fixed-state samplings	presentation/source_data/shot_sensitivity.json
IBM RECORD	job metadata, no counts	output/script_20261107_231254_822277/
REAL NETWORK	CFR + ERA sources	presentation/REAL_ROMANIA_FEASIBILITY.md

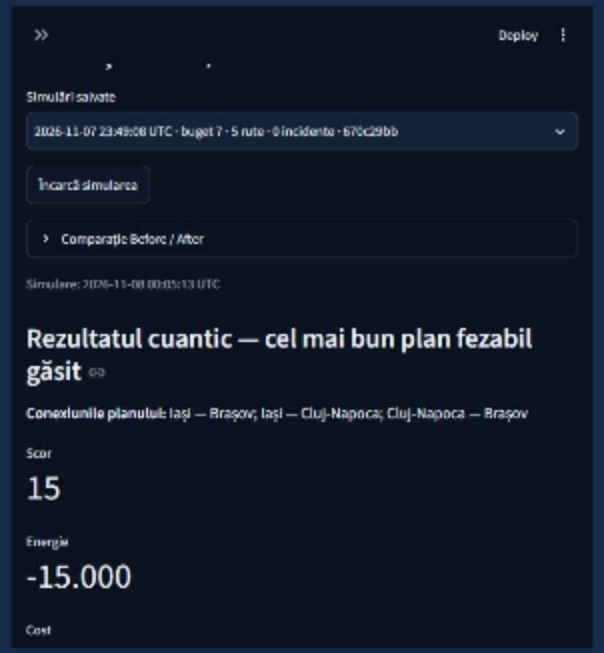
Rebuild: presentation/rebuild.ps1 · IBM read-only refresh: presentation/refresh_qpu_results.py --rebuild

The app presents both quantum and exact plans

Genuine local Streamlit captures from a five-route, budget-seven run.



Main interface



Best feasible sampled



Exact classical

Captured from the locally running app 8 Nov 2026. UI run is separate from frozen experimental JSON.